

Concept mapping as a critical thinking facilitator for generative AI-supported collaborative learning: Effects on diverse perceptions and cognitive levels

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ABSTRACT

Collaborative learning has been identified as an effective means of enhancing students' collaboration and problem-solving skills, as well as boosting their knowledge acquisition. The advent of generative artificial intelligence (GAI) has further provided a potential tool to facilitate diverse thinking in collaborative learning. However, previous studies have revealed that students experience cognitive overload due to the vast amount of available information. To address these issues, it is necessary to provide students with tools or supports for effectively employing GAI in collaborative learning. This study proposed a concept mapping-enhanced GAI-supported collaborative learning (CM-GAI-CL) mode, and evaluated its effectiveness through a quasi-experimental design. Participants were recruited from three high school classes. One class with 33 students served as the experimental group, adopting the CM-GAI-CL mode. Another class with 26 students served as control group 1, using the conventional GAI-supported collaborative learning (C-GAI-CL) mode. The third class with 29 students served as control group 2, using the conventional collaborative learning (C-CL) mode. The results suggested that the experimental group demonstrated significantly better learning achievement, collective efficacy, critical thinking tendencies, and problem-solving skills compared to both control groups.

1. Introduction

Collaborative learning has been recognized as an important and effective approach to fostering students' higher order thinking as well as their collaboration and communication competences (Markula & Aksela, 2022; Quines, 2017; Rorimpandey et al., 2022). In the past decades, several previous studies have demonstrated the effectiveness of engaging students in collaborative learning tasks (Knoche, 2022; Suendarti & Virgana, 2022). For example, in a reading and writing course, Marcos et al. (2020) employed cooperative learning to encourage students to collaboratively analyze and write in groups and develop their own perspectives, and found that this approach effectively enhanced their creative thinking. However, research has indicated that when students engage in group-based problem-solving tasks without sufficient guidance or support, they may feel confused or even experience anxiety during the

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learning process (Tsai & Wu, 2021). When conducting conventional collaborative learning activities, teachers often manage numerous students and groups, making it difficult to adequately address the needs of each group. As indicated by several scholars, collaborative learning facilitates social negotiation and shared knowledge construction; however, it may lack sufficient structure without appropriate scaffolding (Kokku et al., 2018; Tsai & Wu, 2021); that is, providing timely guidance or immediate feedback or guidance to individual collaborative learning groups remains a critical challenge in educational contexts (Hsu et al., 2023; Kokku et al., 2018; Tsai & Wu, 2021).

Generative Artificial Intelligence (GAI) is regarded as a potential solution to address this issue due to its ability to provide diverse and immediate responses, reducing teachers' workloads while offering guidance or suggestions to each group of students during the collaborative learning process (Dwivedi et al., 2023; Fui-Hoon Nah et al., 2023). Nonetheless, scholars have proposed several concerns regarding adopting GAI in learning, including the provision of incorrect information and cognitive overload (Christensen et al., 2024; Zhang et al., 2023). This implies that, during the GAI-supported collaborative learning process, it may be necessary to provide additional scaffolding to help students organize what they have learned from the class and what they have derived from GAI (Hsu et al., 2023; Kasneci et al., 2023; Lin & Mubarak, 2021). In addition, GAI offers informational expansion and immediate feedback, yet it may increase cognitive load or introduce inaccurate content (Dwivedi et al., 2023; Zhang et al., 2023).

To address these complementary limitations, the present study integrated GAI and collaborative learning with a concept-mapping tool as a human-mediated cognitive scaffold. Concept mapping is such a cognitive tool that aids students in visualizing information (Novak, 1983), particularly as a collaborative learning tool for co-constructing knowledge (Shi et al., 2023; Tsourela et al., 2015). Numerous studies have confirmed that concept mapping is an effective tool for promoting meaningful learning (Shi et al., 2023). It helps students identify and evaluate the relationships, significance, and differences between knowledge, thereby enhancing learning efficiency (Evans & Jeong, 2023; Machado & Carvalho, 2020). That is, during the collaborative learning process, concept mapping serves as a metacognitive and organizational scaffold that structures AI-generated information, externalizes learners' reasoning, and supports group regulation (Ma et al., 2023; Su & Zou, 2024). In this study, students were guided to construct concept maps using the designated tool during the GAI-supported collaborative learning process to ensure that knowledge organization reflected learners' own cognitive processing instead of automated outputs.

Grounded in Social Cognitive Theory (Bandura, 1986, 2000), the integration of structured cognitive organization and collaborative regulation is expected to enhance learners' collective efficacy, which in turn promotes strategic engagement in complex tasks. Such strategic and metacognitive engagement fosters the development of higher-order cognitive processes, particularly critical thinking tendencies and problem-solving skills (Barta et al., 2022; Lai & Hwang, 2014). Within this framework, collective efficacy functions as a motivational driver that strengthens engagement, supports higher-order thinking, and ultimately contributes to academic achievement and learning experiences. Improvements in these cognitive and motivational processes are subsequently translated into gains in academic achievement, while learners' perceived learning experiences represent affective-cognitive evaluations emerging from successful task engagement and collaborative functioning (Avry et al., 2020; Fredricks et al., 2004; Schunk & DiBenedetto, 2020).

Thus, drawing upon Krajcik and Shin's (2014) framework of key characteristics of problem-based learning (PBL), this study proposed a concept mapping-enhanced GAI-supported collaborative learning mode (CM-GAI-CL) as the primary instructional intervention. To evaluate its effectiveness, a course unit on Sustainable Development was implemented to compare three instructional conditions: conventional collaborative learning (C-CL), GAI-supported collaborative learning (GAI-CL), and CM-GAI-CL. The instructional mode served as the independent variable, whereas academic achievement, collective efficacy, critical thinking tendencies, problem-solving skills, and learning experiences were examined as dependent variables. The study addressed the following research questions:

- (1) Can the CM-GAI-CL mode improve students' learning achievement compared to the C-CL mode and the GAI-CL mode?
- (2) Can the CM-GAI-CL mode enhance students' collective efficacy compared to the C-CL and GAI-CL modes?
- (3) Can the CM-GAI-CL mode increase students' critical thinking tendencies compared to the C-CL and GAI-CL modes?
- (4) Can the CM-GAI-CL mode enhance students' problem-solving skills compared to the C-CL and GAI-CL modes?
- (5) What are the learning experiences and perceptions of students adopting the CM-GAI-CL and GAI-CL modes?

2. Literature review

2.1. Collaborative learning and problem-based learning (PBL)

In the field of education, collaborative learning has been widely applied and is regarded as an effective instructional strategy for enhancing student learning (Gillies, 2003; Hron & Friedrich, 2003; Kokku et al., 2018). Several researchers have revealed that collaborative learning can significantly improve students' academic performance (Rorimpandey et al., 2022; Suendarti & Virgana, 2022), collective efficacy (Gillies, 2019), critical thinking tendencies (Huang et al., 2017; Quines, 2017), as well as problem-solving and decision-making abilities (Knoche, 2022). These benefits can also be explained by Social Cognitive Theory, which emphasizes that learning occurs through social interaction and observational processes within a shared environment (Bandura, 1986, 2000). Building upon the foundations of collaborative learning, PBL has emerged as a structured approach that embodies its core principles. It is commonly regarded as a concrete form of collaborative learning, and is deeply inspired by the theory of social constructivism (Vygotsky, 1986). According to Vygotsky's (1986) view of social constructivism, knowledge is constructed through social interaction, language communication, and collaborative activities in cultural contexts. In a PBL context, students discuss, reason, and share knowledge about real problems in small groups, and develop understanding and solutions through peer collaboration and reflection

(Markula & Aksela, 2022; Tsai & Wu, 2021). This learning process strengthens learners' active participation and realizes the key role of social interaction in cognitive development and knowledge construction. Krajcik and Shin's (2014) framework of the key characteristics of PBL divides collaborative learning into different stages, namely a driving question, learning goals, scientific practices, collaboration, using technological tools, and creating an artefact, which help to enhance students' problem-solving skills. Previous research has shown that engaging students in collaborative learning and PBL tasks can enhance not only their academic achievement but also their group self-efficacy, critical thinking tendencies, and problem-solving skills (Huang et al., 2017; Knoche, 2022; Ma et al., 2023).

Although the benefits of collaborative learning have been well verified, and approaches such as PBL are grounded in collaborative principles, simply grouping students and asking them to work together does not necessarily ensure effective collaboration or positive learning outcomes (Boud & Bearman, 2024; Gillies, 2003). Researchers have noted that when students engage in group-based problem-solving tasks without expert guidance, they may feel confused or even experience anxiety (Tsai & Wu, 2021). In traditional teaching environments, providing timely and individualized guidance can be challenging for educators (Hsu et al., 2023). Hence, implementing effective scaffolding strategies and designing appropriate learning tasks are essential for creating a supportive environment that facilitates discussion and learning (Hron & Friedrich, 2003). In recent years, some researchers have suggested employing GAI systems, such as ChatGPT, as virtual mentors to support collaborative learning (Fui-Hoon Nah et al., 2023; Lai & Tu, 2024; Weng & Chiu, 2023).

2.2. GAI in collaborative learning

GAI, an AI technology capable of creating new content (e.g., text, images, or audio) based on input data, has recently become an important trend in the field of education (Chiu, 2023; Dwivedi et al., 2023). Large Language Models (LLMs) represent a core technological foundation of GAI in the domain of natural language processing, enabling the automated generation of coherent, contextually appropriate text based on extensive training on linguistic data (Kasneci et al., 2023). For example, ChatGPT, with its powerful language understanding and generation capabilities, can provide real-time interaction and question answering during the learning process, effectively assisting students with their knowledge construction (Jeon et al., 2023; Michel-Villarreal et al., 2023).

From the perspective of social constructivism, which emphasizes the co-construction of knowledge through social interaction, collaboration, and situated learning within meaningful contexts, the integration of GAI can enhance the diversity and depth of group discussions, thereby fostering students' critical thinking and problem-solving abilities (Kasneci et al., 2023; Rudolph et al., 2023). Concept mapping can be understood as a metacognitive scaffolding tool that supports learners' cognitive organization and self-regulation processes (Tsourela et al., 2015). From the perspective of Social Cognitive Theory (Bandura, 1986, 2000), such external cognitive supports may facilitate learners' strategic engagement with complex tasks by helping them monitor, structure, and evaluate their knowledge construction processes. When integrated with collaborative learning, concept mapping may also enhance collective efficacy by enabling groups to externalize and coordinate their shared understanding. The extensive knowledge and diverse perspectives provided by GAI contribute to broadening the scope of discussions, encouraging students to engage in multifaceted thinking from various viewpoints. These diverse perspectives can then be incorporated into group discussions and problem-solving processes, thereby facilitating collaborative learning activities (Chen et al., 2023; Dwivedi et al., 2023; Neto et al., 2019). When students understand how they should work together, the potential for collaboration and learning is maximized (Gillies, 2003; Hron & Friedrich, 2003; Johnson & Johnson, 2009).

Although GAI can generate convincing narratives, previous studies have pointed out that it suffers from the problem of "hallucination," that is, it generates information that does not correspond to the truth, but is linguistically and logically fluent, which is very likely to mislead learners (Lee et al., 2024; Zhang et al., 2023). For example, when asked about the social impact of sunset industries, GAI may fabricate non-existent industrial policies, misquote inaccurate cases, or misinterpret historical facts. This may reinforce students' misperceptions if they do not have sufficient background knowledge or critical validation skills. Some studies have raised concerns regarding the application of GAI, particularly regarding the accuracy and reliability of the information it generates, as well as its tendency to produce both relevant and irrelevant content (Dwivedi et al., 2023; Michel-Villarreal et al., 2023). Consequently, it is crucial to support students in effectively integrating the knowledge they acquire, as mere knowledge acquisition is insufficient. The key lies in how students integrate, apply, and transform their knowledge (Machado & Carvalho, 2020; Shi et al., 2023). Otherwise, engaging in discussions and interactions with GAI may fail to produce meaningful learning outcomes, and could even impose an additional cognitive load on students. Therefore, providing appropriate learning scaffolding is essential when utilizing GAI for learning, as this improves the efficiency and effectiveness of students' interactions with AI (Kasneci et al., 2023; Lin & Mubarak, 2021).

2.3. Concept mapping as metacognitive scaffolding

Concept mapping is a knowledge organization tool consisting of nodes and connecting lines, where nodes represent concepts, and lines indicate the relationships between them (Novak, 1983). The use of concept mapping extends across various domains. In workplace management, it can be employed for strategic planning, problem solving, and idea communication (Yue et al., 2017). As for language education, concept mapping is used to organize and present complex textual content, as well as to support instructional strategies such as map completion and collaborative scaffolding (Su & Zou, 2024). Regarding mathematics education, teachers can utilize concept mapping to assist students in better understanding course content, thereby enhancing knowledge retention and application (Evans & Jeong, 2023). For instance, Lin et al. (2022) incorporated concept mapping into nursing courses to assist students in organizing and analyzing information, facilitating the integration of multidisciplinary theoretical knowledge. The results indicated

that the experimental group demonstrated significantly higher academic performance and self-efficacy. Above all, the collaborative construction of concept mapping can serve as an effective approach to promoting learners' knowledge construction (Shi et al., 2023; Tsourela et al., 2015).

Moreover, when students encounter large amounts of information during the learning process, concept mapping can facilitate the organization, comprehension, and application of the acquired information. This is particularly beneficial for collaborative learning when students need to process substantial amounts of information, explore complex topics, or solve problems (Ma et al., 2023; Tsourela et al., 2015). Such a visual tool provides both structural and visual support (Novak, 1983), enabling students to more effectively utilize the extensive information offered by GAI.

Scholars have further indicated that concept mapping can serve as a metacognitive scaffolding tool, helping to reduce cognitive load, structure group interactions, and enhance the effectiveness of collaborative problem solving (Machado & Carvalho, 2020; Su & Zou, 2024). For example, in GAI-supported learning, concept maps can function as a shared external representation that helps learners formulate more purposeful prompts and questions, and critically evaluate information returned by the chatbot, thereby improving AI student interaction quality (Ye et al., 2025). Concept maps make learners' conceptual structure explicit and revisable, prompting planning (what to include), monitoring (whether links are coherent), and evaluation (what is missing or unsupported) during knowledge construction (Novak, 1990). The cooperative learning literature further emphasizes that positive interdependence, individual accountability, promotive interaction, interpersonal and small-group skills, and group processing are essential dimensions that support effective collaborative learning outcomes (Johnson & Johnson, 2009; Machado & Carvalho, 2020). Aligning concept mapping with these cooperative-learning dimensions strengthens the theoretical rationale for its use in GAI-supported collaborative learning, because the shared map can operationalize interdependence and promote interaction while providing a concrete basis for accountability and group processing.

From the cognitive load perspective, concept maps act as external representations that offload working-memory demands and stabilize complex relations so that learners can allocate more resources to schema construction and meaningful integration, rather than to keeping information "in mind" (Kirsh, 2010; Sweller, 1988). This same externalization can enhance AI-student interaction by providing a shared, persistent artifact that constrains prompts, reduces ambiguity, and supports iterative cycles of querying, critiquing, and revising (Azevedo & Hadwin, 2005; Kirsh, 2010). Research on AI-supported scaffolding has suggested that structured scaffolds paired with AI agents can improve comprehension and learning processes relative to unstructured use, which supports the design rationale for using concept maps as a prompt scaffold, rather than treating AI interaction as ad hoc questioning (Yan et al., 2025).

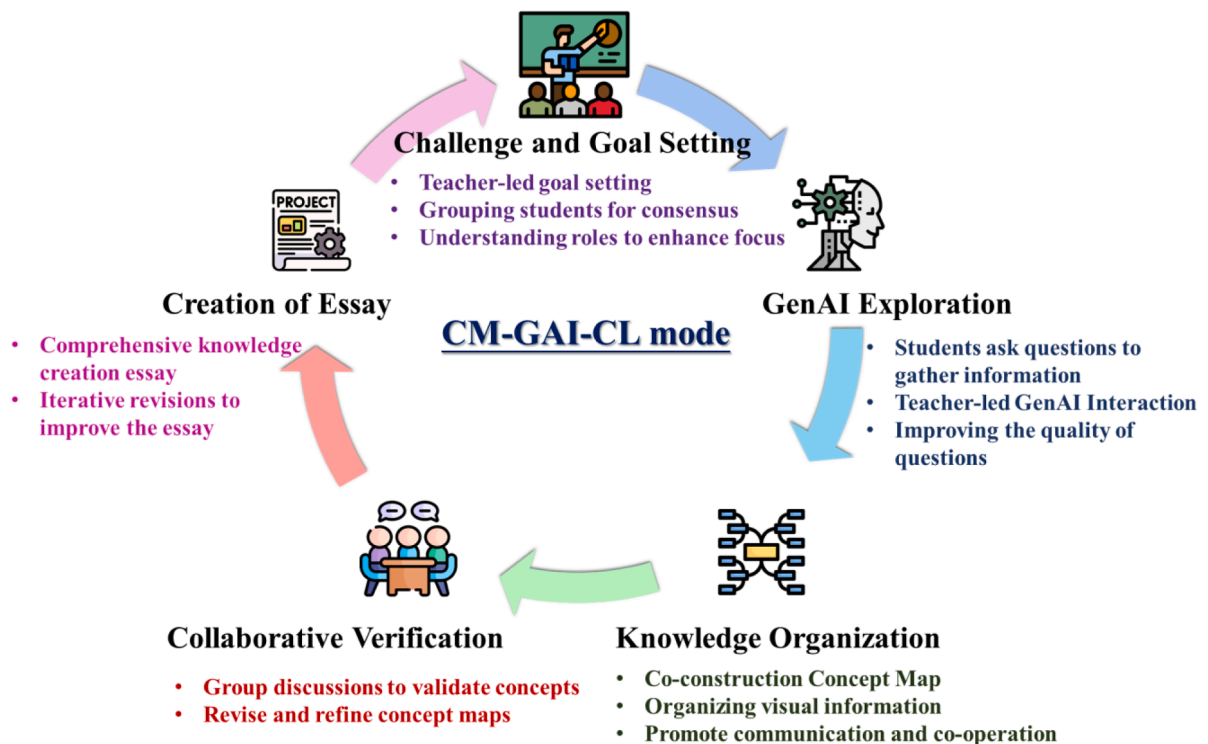


Fig. 1. Stages of the CM-GAI-CL mode.

3. Methodology

3.1. Participants

The participants in this study were second-year high school students (11th-grade students) from a school in central Taiwan, with an average age of 17. All participants, representing three classes, volunteered to take part in the study. In adherence to ethical guidelines stipulated by the Institutional Review Board, all participant data were anonymized, and students could withdraw from the study at any time without providing a reason. One class of 33 students served as the experimental group, another class of 26 students was assigned as control group 1, and the other class of 29 students was control group 2. The experimental group adopted the CM-GAI-CL mode, control group 1 employed the GAI-CL mode, and control group 2 utilized the C-CL mode. Previous studies have demonstrated that the optimal group size for collaborative learning is three students per group (Beghetto, 2016; Gillies, 2019; Hsu et al., 2009). Free grouping was adopted by referring to the suggestions of Hsu et al. (2009) and Hu et al. (2022), with three students in each group. This allowed students to form teams according to their familiarity or learning styles so as to promote natural cooperative interactions and discussion quality. None of the participants had prior learning experience of the course unit. A pretest was administered to ensure that all students had comparable prior knowledge related to sustainability before the experiment.

3.2. Concept mapping-enhanced GAI-supported collaborative learning (CM-GAI-CL) model

This study builds on Krajcik and Shin's (2014) framework of the key characteristics of PBL. As illustrated in Fig. 1, the proposed CM-GAI-CL model includes five progressive stages designed not only to support academic development but also to foster students' abilities to engage critically with sustainability issues, understand systemic complexity, and collaborate toward socially responsible solutions.

This study was conducted within the context of a course unit regarding a sustainable development issue, namely "sunset industries." The topic was deliberately selected as a real-world, interdisciplinary sustainability challenge. It engages students in analyzing the socio-economic and environmental impacts of industrial decline and technological transformation. Through this issue, students examine how economic restructuring intersects with labor rights, ecological concerns, and policy-making, thereby promoting systems thinking. Concept mapping played a central role by helping students visualize the dynamic interconnections among technological advancement, economic policy, employment shifts, and cultural values. This process supported the construction of holistic, interconnected understandings of sustainability problems.

In the "Challenge and Goal Setting" stage, the teacher posed problem-based questions and guided students in understanding the learning objectives, ensuring a clear direction for the learning activity. According to Locke and Latham's (2002) goal-setting theory, well-defined goals can enhance learners' motivation and focus. The teacher had to not only convey information, but also guide students to examine the essence and underlying significance of the problem, fostering a comprehensive understanding of the core learning tasks. Subsequently, with the teacher's guidance, students participated in group discussions to develop specific and achievable learning objectives. This process promoted active engagement, enabling students to share their perspectives and insights and reach a consensus. Through this collaborative effort, students gained a clear understanding of their roles and responsibilities in the learning process, thereby improving their focus and learning effectiveness. This enabled them to prepare to explore and construct knowledge by interacting with GAI in the subsequent stages.

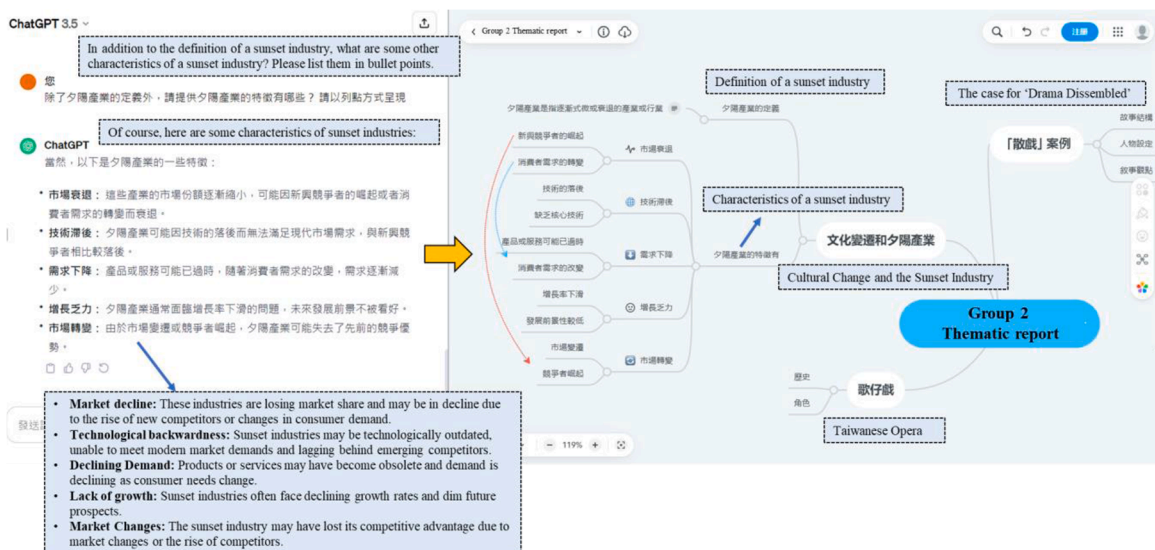


Fig. 2. Knowledge organization: a group constructed knowledge through concept mapping.

In the “GAI Exploration” stage, students interacted with GAI to gather data and explore knowledge and information related to the questions. Throughout this process, the teacher guided students in effectively engaging with GAI through a series of steps: defining questions, requesting details for clarification, developing ideas, drilling down with follow-up inquiries, and verifying the accuracy of the collected information to make necessary adjustments. According to recommendations from previous studies (Banh & Strobel; Fui-Hoon Nah et al., 2023; Hsu et al., 2023), GAI is an emerging technological tool that requires users to follow specific guidelines for more effective interaction. In the example of the “sunset industry” issue, students might ask: “What are the characteristics of a sunset industry?” and “Could you give examples of sunset industries facing labor or environmental issues?” GAI not only provided abstract concepts such as market decline, technological backwardness, and lack of growth, but also supported them with explanations and concrete terms. Students were advised to make preliminary judgments on the correctness of the information provided by GAI. These interactions could help students develop skills of critical thinking and effectively asking questions, especially by evaluating the diverse viewpoints offered by GAI.

In the “Knowledge Organization” stage, each group collaboratively used MindMaster to construct a concept map that organized the information gathered from GAI (see Fig. 2). MindMaster was selected for its collaborative features, specifically allowing multiple students to simultaneously edit and visualize concept maps in real time. Its limitations include reliance on stable internet access and a short learning curve for students unfamiliar with digital mapping tools. The students were guided to check the information from GAI and identify the relationships between the concepts underlying the information, such as “characteristics of sunset industries,” “labor protection,” and “environmental sustainability.” They then visualized the relationships by developing concept maps. The process of constructing a concept map was not only an integration of knowledge but also a test of students’ depth of understanding. This not only aided their own comprehension and memory, but also facilitated communication and discussion among group members. Therefore, constructing a concept map further promoted student communication and collaboration, ensuring that each member actively participated in knowledge construction.

In the “Collaborative Verification” stage, students shared and discussed the information and perspectives they obtained from GAI within their group. They engaged in collaborative learning through the constructed concept map to verify the accuracy of concepts and make revisions accordingly (see Fig. 3). This stage focused on verifying information and reflection content in the group learning. The students were encouraged to add real-life examples (e.g., traditional retail vs. e-commerce, print publishers vs. digital media publishers), strategies to prevent sunset industries from disappearing (e.g., digital transformation, improving product or service innovation and repositioning the market), and information about their impact on the economy (e.g., employment impact, slow economic growth and supply chain impact) to their concept maps. This content comprised the learning outcomes inspired by students’ interactions with GAI and discussions with classmates. This process aligned with Vygotsky’s (1986) social constructivism, which emphasizes the promotion of knowledge internalization and construction through social interaction and collaborative learning. Students used MindMaster to display and refine their concept maps, documenting the outcomes and improvement suggestions from group discussions. Meanwhile, GAI addressed students’ advanced questions to help them add information. In addition, students continued to revise and refine their concept maps through group “collaboration,” “reflection,” and “validation.” Such processes helped the students optimize their learning outcomes by further deepening their understanding and confirming the accuracy of the information.

Finally, in the “Creation of an Essay” stage, students created a written essay based on their acquired knowledge and inquiry

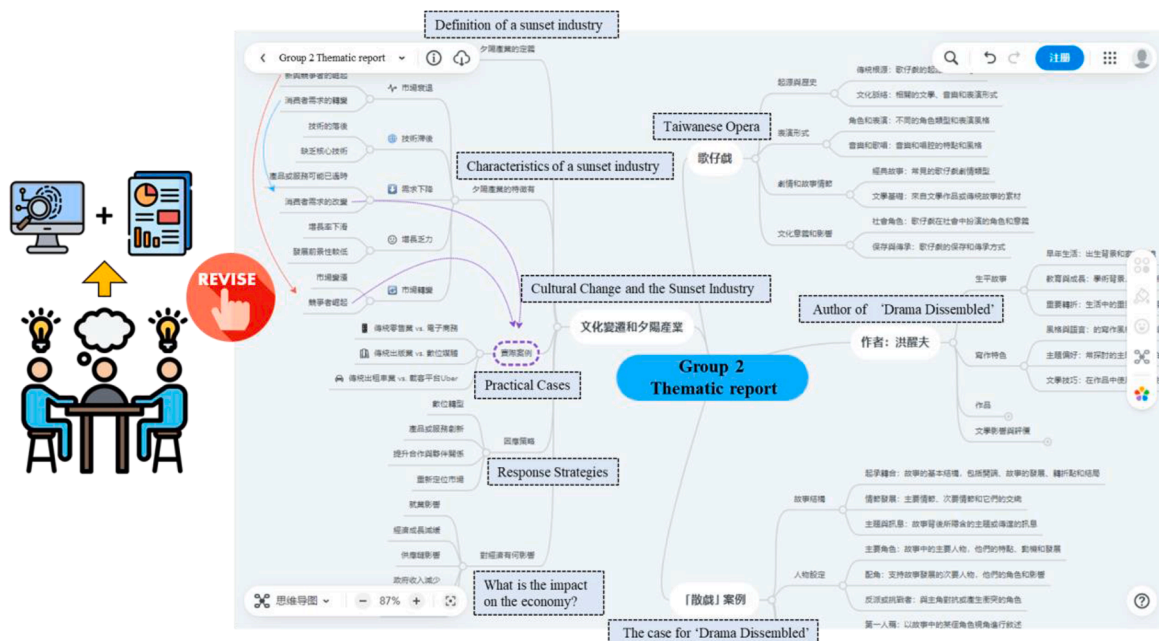


Fig. 3. Collaborative verification: a group collaborated to revise the concept map.

findings, including conversations with GAI and the information from their concept maps. During this process, GAI assisted students in planning and formatting the essay, offering suggestions on language, organization, and structure. Additionally, students were required to integrate the knowledge they had gained, and to engage in iterative revisions until the final essay was completed. This process served not only as a summary of knowledge, but also as a comprehensive assessment of students' learning and collaboration skills. The iterative revisions involved in the written essay aligned with the highest level of Bloom's (1956) taxonomy of cognitive learning objectives, namely Creating, which requires students to synthesize and apply their acquired knowledge in innovative ways and express their ideas. The essay also helped students organize and summarize their learning, while enhancing their writing and communication skills (see Fig. 4).

3.3. Experimental procedure

The experimental procedure is shown in Fig. 5. Prior to the learning activity, all three groups completed a 60-min pretest and pre-questionnaires on collective efficacy, critical thinking tendencies, and problem-solving skills during the first week. All questionnaires used in this study were well-recognized instruments published in academic journals, and had undergone reliability and validity testing to ensure the accuracy and credibility of the measurements.

Over the following 3 weeks, all students participated in structured 100-min weekly sessions, totaling 300 min, working together within their groups to complete their projects. Students engaged in collaborative learning to produce a "sunset industries" essay, which they submitted to the designated platform by the specified deadline.

The experimental group received training on utilizing concept mapping and GAI (i.e., ChatGPT 3.5 version), adopting the CM-GAI-CL mode for their learning activity. Control group 1 received training on using GAI and employed the GAI-CL mode, while control group 2 was trained in the use of online search tools and adopted the C-CL mode for their learning. Throughout the learning process, all three groups adhered to Krajcik and Shin's (2014) framework of the key characteristics of PBL. The experimental group, adopting the CM-GAI-CL mode, utilized answers provided by ChatGPT to understand the relationships between concepts and to improve their concept maps. Finally, students verified the information and made modifications or adjustments to their concept maps. The primary distinction between the experimental group and control group 1 was the use of concept mapping. In contrast, control group 2 employed a conventional approach, relying on online information searches. During the learning process, students in control group 2 were introduced to commonly used search engines such as Google, and were instructed to use these tools to search for relevant learning materials. They were guided to document useful sources, preliminarily organize the information, and verify its accuracy in order to complete their written essays. Additionally, all three groups were required to document the division of tasks in their essays to ensure that each student's contribution was clearly outlined.

Except for the differences in learning modes, the learning content, instructional materials, resources for the written essay, and assessment procedures were identical across all groups to control for teacher and classroom effects. The same teacher implemented all instructional conditions following a standardized teaching procedure. Standard prompts for ChatGPT-3.5 were derived from a teaching manual, and were administered consistently across all three groups by the same instructor to ensure uniformity in students' interactions with the AI. Some example prompts are "What are the main causes and impacts of this sunset industry?" "Can you propose possible solutions or transformation strategies based on sustainability principles?" and "What additional information or perspectives should be considered when analyzing this issue?" Students were allowed to generate follow-up questions collaboratively; however, all initial prompts and task instructions were standardized across groups. Throughout the intervention, the teacher monitored students'



Fig. 4. Creation of an essay: a group collaborated to complete the essay.

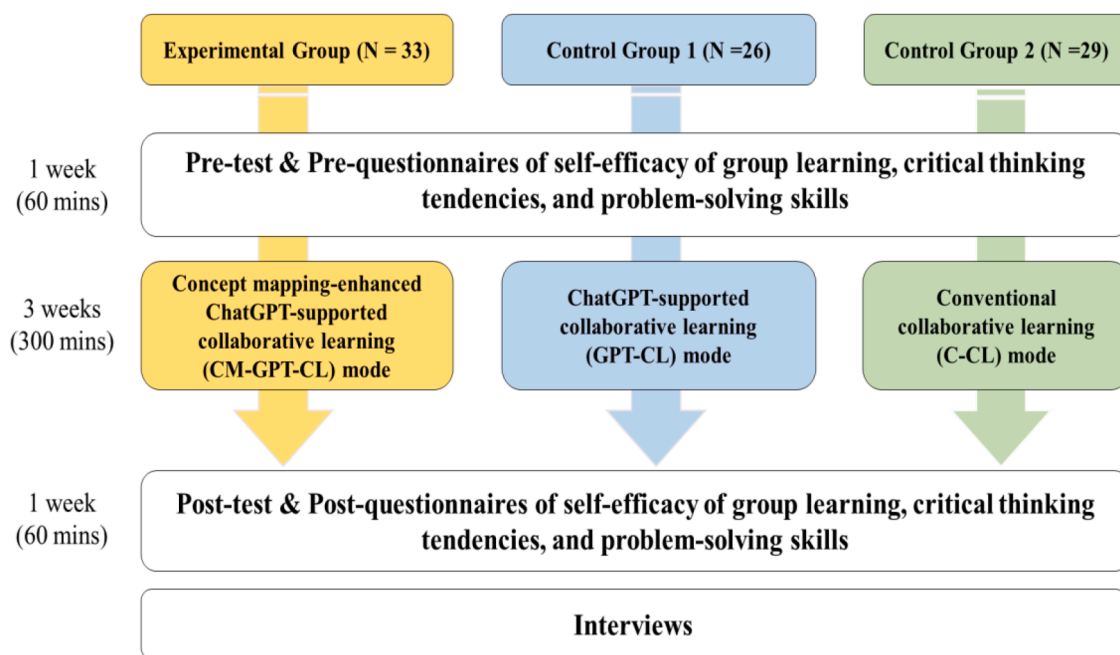


Fig. 5. Experimental procedure.

learning processes, provided guidance when necessary, clarified conceptual misunderstandings, and recorded students' engagement behaviors for further analysis. After completing the essays, all three groups undertook a 60-min posttest and completed post-questionnaires, followed by interviews.

3.4. Measuring tools

This study employed five measurement tools: a learning achievement test, a collective efficacy questionnaire, a critical thinking tendencies questionnaire, a problem-solving skills questionnaire, and interview questions.

The pretest aimed to assess students' prior knowledge, while the posttest aimed to evaluate their learning achievement after the instructional intervention. Both tests consisted of 36 multiple-choice items developed based on Bloom's (1956) taxonomy of cognitive learning objectives, including 20 "remembering" items (30%), 10 "understanding" items (45%), and six "applying and analyzing" items (25%). Each test had a total score of 100. The pretest and posttest demonstrated acceptable reliability, with KR-20 values of 0.74 and 0.79, respectively. The test items and learning materials were collaboratively developed by two high school teachers and educational experts, each with >10 years of teaching experience, to ensure content validity and clarity.

The collective efficacy questionnaire was adopted from Lai and Hwang (2014) and Wang and Lin (2007)). It measures students' perceived collective capability in collaborative learning contexts. The scale was originally developed by Lai and Hwang (2014), who reported acceptable construct validity using confirmatory factor analysis (CFA). It included eight items, for example, "I believe that through collaborative learning, I can understand the most complex concepts taught in this course."

The critical thinking tendencies questionnaire used in this study was adopted from Chai et al. (2015), who adapted the original instrument developed by Castle (2006). Because the Chai et al. (2015) version was developed for information communication technology (ICT)-supported learning environments, it was considered more appropriate for the context of GAI-supported learning in the present study. The validated six-item version was used, including items such as, "In the learning process, I evaluate the value of new information or evidence presented to me." After translation into Chinese, the questionnaire was reviewed by two experts to confirm the clarity and appropriateness of the translated items for 11th-grade students. Minor wording revisions were made without altering the

Table 1
Cronbach's alpha coefficients for each questionnaire at the pretest and posttest.

Measure	Stage	No. of items	Reliability
Collective efficacy	Pretest	8	0.92
	Posttest	8	0.91
Critical thinking tendencies	Pretest	6	0.93
	Posttest	6	0.86
Problem-solving skills	Pretest	6	0.91
	Posttest	6	0.91

underlying construct. In addition, this instrument has previously been used in studies involving high school students (e.g., Chien et al., 2020; Huang et al., 2026), and it demonstrated satisfactory internal consistency in the present study (Cronbach's $\alpha = 0.93$ for the pretest and 0.86 for the posttest).

The problem-solving skills questionnaire was adopted from Lai and Hwang (2014). It evaluates students' strategic approaches and confidence in resolving learning-related problems. The scale was originally developed by Lai and Hwang (2014), who reported acceptable construct validity using CFA. It consisted of six items, for example, "I always do my best to solve the problems I encounter."

All instruments were directly translated from their original versions. To ensure content validity and contextual appropriateness, all items were reviewed by two experts in educational psychology and collaborative learning, after which minor wording revisions were made accordingly. In addition, all questionnaires adopted a 5-point Likert scale, and Cronbach's alpha coefficients for both measurement points (pretest and posttest) are presented in Table 1.

The eight interview questions were adapted from Hwang et al. (2009) and were modified based on the CM-GAI-CL mode for this study. The aim was to gain insights from students' open-ended responses regarding their perspectives on the different learning modes. Sample questions include: "How does this learning mode differ from the approaches you have experienced (or expected) before?" and "What did you gain the most from using this mode? What did you learn the most?"

For the interview data, the coding procedures, coder training, intercoder reliability, and determination of data saturation were carefully documented to ensure transparency and rigor in the qualitative analysis. In addition to reporting internal consistency reliability, we clarified the original sources and theoretical grounding of each instrument to strengthen methodological transparency.

4. Experimental results

The data for learning achievement, collective efficacy, critical thinking tendency, and problem-solving skills were found to meet the assumptions of normal distribution. To address the first research question, the internal consistency of the pre- and posttests was first assessed using KR20. Subsequently, an analysis of covariance (ANCOVA) was performed to control the differences in pretest scores and to examine whether there was a difference in students' achievements before and after the learning activity.

For the second to fourth research questions, Cronbach's alpha values were first used to assess the reliability of the pre- and post-questionnaires. ANCOVA was then conducted, controlling for the differences in pre-questionnaire scores, to examine whether there were differences in students' collective efficacy, critical thinking tendencies, and problem-solving skills before and after the learning activity.

Finally, for the qualitative data related to the fifth research question, all interview recordings were transcribed verbatim and were coded by two researchers based on the transcripts. Following Creswell's (2013) guidelines, the researchers first conducted initial coding independently. After discussion and negotiation, they established unified coding criteria. Consistency checks revealed a kappa value of 0.81 between the two researchers, indicating high consistency (Bakeman & Gottman, 1997). Any discrepancies during the coding process were resolved through discussion until a consensus was reached.

4.1. Learning achievement

To ensure group equivalence of learning achievement before the experiment, the one-way ANOVA was conducted to compare the pretest scores of the three groups, and no significant difference was found, with $F = 0.51$ ($p > .05$).

To meet the assumptions of the one-way ANCOVA, the Levene's test for determining the homogeneity of variance was conducted. The results indicated that the assumption of homogeneity of variances was not violated ($F = 0.67$, $p = .52 > 0.05$). Additionally, the homogeneity test of the within-group regression coefficients showed that the assumption was met ($F = 0.38$, $p = .69 > 0.05$), suggesting that the linear relationship between the covariate and the dependent variable was consistent across groups. Therefore, ANCOVA could be performed to examine the differences in learning achievement among the three groups. The results are presented in Table 2.

The adjusted mean scores for the experimental group, control group 1, and control group 2 were 64.56, 52.41, and 56.89, respectively. After accounting for the influence of the pretest scores, a significant difference in learning achievement was found among the three groups ($F = 9.47$, $p = .00 < 0.001$, $\eta^2 = 0.18$). The least significant difference (LSD) post hoc tests further revealed that the experimental group outperformed both control groups 1 and 2 after the learning activity, suggesting that the CM-GAI-CL mode contributed to enhanced learning achievement in a collaborative learning environment.

To further explore the learning achievement of the experimental group, control group 1, and control group 2, an additional analysis was conducted using one-way ANCOVA, focusing on the three item types: remembering, understanding, and applying and analyzing.

The results are shown in Table 3. For the "remembering" items, the adjusted mean scores for the experimental group, control group

Table 2
ANCOVA results of learning achievement.

Group	N	Mean	SD	Adjusted Mean	F	p	η^2	Post Hoc (LSD)
(E) Experimental	33	64.22	11.06	64.56	9.47	$p < .001$	0.18	E > C1
(C1) Control 1	26	52.67	15.51	52.41				E > C2
(C2) Control 2	29	56.33	10.51	56.89				

Table 3
ANCOVA results of learning achievement by item types.

	Group	N	Mean	SD	Adjusted Mean	F	p	η^2	Post Hoc (LSD)
Remembering	(E) Experimental	33	21.15	4.61	21.74	10.49	$p < .001$	0.27	E > C1 C2 > C1
	(C1) Control 1	26	16.58	4.82	16.91				
	(C2) Control 2	29	22.86	3.97	21.89				
Understanding	(E) Experimental	33	30.75	6.08	29.18	2.05	0.14	0.14	
	(C1) Control 1	26	26.13	8.44	25.44				
	(C2) Control 2	29	26.22	8.74	28.64				
Applying and analyzing	(E) Experimental	33	13.03	5.10	13.13	13.37	$p < .001$	0.24	E > C2 E > C1
	(C1) Control 1	26	9.96	5.71	9.63				
	(C2) Control 2	29	7.14	2.80	7.32				

1, and control group 2 were 21.74, 16.91, and 21.89, respectively. The scores for remembering differed significantly across the learning modes ($F = 10.49$, $p = .00 < 0.001$, $\eta^2 = 0.27$). The LSD post hoc test revealed that both the experimental group and control group 2 scored significantly higher than control group 1 after the learning activity. This finding suggests that in a collaborative learning environment, both the CM-GAI-CL mode and the C-CL mode effectively enhanced students' learning achievement on the "remembering" items. As for the "understanding" items, the adjusted mean scores for the experimental group, control group 1, and control group 2 were 29.18, 25.44, and 28.64, respectively. No significant difference in scores for understanding across the learning modes could be found ($F = 2.05$, $p = .14 > 0.05$). Regarding the "applying and analyzing" items, the adjusted mean scores for the experimental group, control group 1, and control group 2 were 13.13, 9.63, and 7.32, respectively. There was a significant difference in the scores for applying and analyzing across the learning modes ($F = 13.37$, $p = .00 < 0.001$, $\eta^2 = 0.24$). The LSD post hoc test revealed

Table 4
Content analysis of the "applying and analyzing" items.

Category/Group	(E) Experimental	(C1) Control 1	(C2) Control 2
Revising the sunset industry they selected after the group discussion	11	6	4
A1 Final selection of the sunset industry	A1-01:9 A1-02:0 A1-03:7 A1-04:1 A1-05:4 A1-06:8 A1-07:0 A1-08:4	A1-01:8 A1-02:0 A1-03:6 A1-04:0 A1-05:1 A1-06:7 A1-07:3 A1-08:1	A1-01:4 A1-02:2 A1-03:13 A1-04:1 A1-05:1 A1-06:3 A1-07:5 A1-08:0
A2 Reasons for the disappearance of the industry	A2-01:8 A2-02:17 A2-03:1 A2-04:5 A2-05:3 A2-06:10 A2-07:9 A2-08:7 A2-09:4 A2-10:4 Total: 68	A2-01:3 A2-02:10 A2-03:0 A2-04:1 A2-05:0 A2-06:1 A2-07:4 A2-08:6 A2-09:0 A2-10:0 Total: 25	A2-01:2 A2-02:12 A2-03:5 A2-04:6 A2-05:3 A2-06:6 A2-07:4 A2-08:1 A2-09:0 A2-10:0 Total: 39
A3 Measures to improve or delay the disappearance of the industry	A3-01:13 A3-02:5 A3-03:4 A3-04:2 A3-05:7 A3-06:15 A3-07:0 A3-08:2 A3-09:1 A3-10:13 A3-11:2 Total: 64	A3-01:4 A3-02:4 A3-03:1 A3-04:1 A3-05:2 A3-06:7 A3-07:1 A3-08:1 A3-09:1 A3-10:0 A3-11:0 Total: 22	A3-01:12 A3-02:3 A3-03:3 A3-04:2 A3-05:7 A3-06:4 A3-07:1 A3-08:0 A3-09:0 A3-10:0 A3-11:0 Total: 32
A4 New facts or perspectives learned in group discussions	A4-01:14 A4-02:13 A4-03:5 A4-04:10 A4-05:0 A4-06:1 A4-07:2 Total: 45	A4-01:4 A4-02:5 A4-03:4 A4-04:3 A4-05:0 A4-06:0 A4-07:0 Total: 16	A4-01:1 A4-02:2 A4-03:3 A4-04:4 A4-05:5 A4-06:0 A4-07:0 Total: 15

that the experimental group significantly outperformed control groups 1 and 2 after the learning activity. This finding implies that the CM-GAI-CL mode was effective in terms of improving students' learning achievement on the "applying and analyzing" items.

To understand students' learning conditions related to the "applying and analyzing" items, we conducted a content analysis based on the coding scheme (see Appendix A). The results are illustrated in Table 4. After the learning activity, 11 students (33%) from the experimental group, six (23%) from control group 1, and four (14%) from control group 2 revised the sunset industry they selected. The bolded text represents the top three sub-categories for each group, while the underlined text indicates the sub-category for which only one group of students answered. The analysis revealed that for the "A2 Reasons for the disappearance of the industry" category, the experimental group identified 68 reasons (an average of 2.1 per student), while control group 2 identified 39 reasons (an average of 1.3 per student) and control group 1 identified 25 reasons (an average of 1.0 per student). A similar trend was observed in the "A3 Measures to improve or delay the disappearance of the industry" category. The experimental group provided 64 measures (an average of 1.9 per student), while control group 2 provided 32 measures (an average of 1.2 per student) and control group 1 provided 22 measures (an average of 0.8 per student). In terms of the "A4 New facts or perspectives learned in group discussions" category, the experimental group identified 45 points (an average of 1.4 per student), while control group 1 identified 16 points (an average of 0.6 per student) and control group 2 identified 15 points (an average of 0.5 per student).

4.2. Collective efficacy

To ensure group equivalence of collective efficacy before the experiment, a one-way ANOVA was conducted to compare the pretest scores of the three groups, with no significant difference found, $F = 0.70$ ($p > .05$).

To meet the assumptions of the one-way ANCOVA, the Levene's test of determining the homogeneity of variance was performed. The results revealed that the assumption of homogeneity of variances was met ($F = 2.06$, $p = .13 > 0.05$). Moreover, the homogeneity test of the within-group regression coefficients showed that the assumption was not violated ($F = 1.67$, $p = .20 > 0.05$), implying that the linear relationship between the covariate and the dependent variable was consistent across groups. Thus, ANCOVA could be conducted to explore the differences in collective efficacy among the three groups. The results are presented in Table 5.

The adjusted mean scores for the experimental group, control group 1, and control group 2 were 4.08, 3.77, and 3.54, respectively. After accounting for the influence of the pre-questionnaire scores, a significant difference in collective efficacy was found among the three groups ($F = 6.56$, $p = .002 < 0.005$, $\eta^2 = 0.14$). The LSD post hoc tests indicated that the experimental group outperformed both control groups 1 and 2 after the learning activity, suggesting that the CM-GAI-CL mode effectively improved students' collective efficacy in a collaborative learning environment.

4.3. Critical thinking tendencies

To ensure group equivalence of critical thinking tendencies before the experiment, a one-way ANOVA was conducted to compare the pretest scores of the three groups, and a significant difference was found with $F = 5.3$ ($p < .05$). Following that, ANCOVA was employed to control for the influence of pretest scores when comparing posttest outcomes.

To meet the assumptions of the one-way ANCOVA, the Levene's test of determining the homogeneity of variance was conducted. The results indicated that the assumption of homogeneity of variances was not violated ($F = 1.63$, $p = .20 > 0.05$). Additionally, the homogeneity test of the within-group regression coefficients showed that the assumption was met ($F = 1.18$, $p = .31 > 0.05$), suggesting that the linear relationship between the covariate and the dependent variable was consistent across groups. Therefore, ANCOVA could be performed to investigate the differences in critical thinking tendencies among the three groups. The results are presented in Table 6.

The adjusted mean scores for the experimental group, control group 1, and control group 2 were 4.34, 4.08, and 3.93, respectively. After accounting for the influence of the pre-questionnaire scores, a significant difference in critical thinking tendencies was found among the three groups ($F = 4.44$, $p = .015 < 0.05$, $\eta^2 = 0.10$). The LSD post hoc tests revealed that the experimental group outperformed both control groups 1 and 2 after the learning activity, showing that the CM-GAI-CL mode effectively improved students' critical thinking tendencies in a collaborative learning environment.

4.4. Problem-solving skills

To ensure group equivalence of problem-solving skills before the experiment, a one-way ANOVA was conducted to compare the pretest scores of the three groups, and no significant difference was found, with $F = 1.3$ ($p > .05$).

To meet the assumptions of the one-way ANCOVA, the Levene's test of determining the homogeneity of variance was performed.

Table 5
ANCOVA results of collective efficacy.

Group	N	Mean	SD	Adjusted Mean	F	p	η^2	Post Hoc (LSD)
(E) Experimental	33	4.08	0.44	4.06	6.56	$p < .001$	0.14	E > C2
(C1) Control 1	26	3.77	0.68	3.75				E > C1
(C2) Control 2	29	3.54	0.64	3.60				

Table 6
ANCOVA results of critical thinking tendencies.

Group	N	Mean	SD	Adjusted Mean	F	p	η^2	Post Hoc (LSD)
(E) Experimental	33	4.34	0.34	4.32	4.44	$p < .001$	0.10	E > C2
(C1) Control 1	26	4.08	0.44	4.06				E > C1
(C2) Control 2	29	3.93	0.59	3.98				

The results revealed that the assumption of homogeneity of variances was met ($F = 2.26, p = .20 > 0.11$). Moreover, the homogeneity test of the within-group regression coefficients showed that the assumption was not violated ($F = 1.24, p = .29 > 0.05$), implying that the linear relationship between the covariate and the dependent variable was consistent across groups. Thus, ANCOVA could be conducted to explore the differences in problem-solving skills among the three groups. The results are presented in Table 7.

The adjusted mean scores for the experimental group and for control groups 1 and 2 were 4.34, 4.08, and 3.93, respectively. After accounting for the influence of the pre-questionnaire scores, a significant difference in problem-solving skills was found among the three groups ($F = 5.18, p = .008 < 0.01, \eta^2 = 0.10$). The LSD post hoc tests indicated that the experimental group outperformed both control groups 1 and 2 after the learning activity, implying that the CM-GAI-CL mode effectively enhanced students' problem-solving skills in a collaborative learning environment.

4.5. Interview results

After the experiment, interviews were conducted with each group of students to gather their perspectives on the learning activity. The feedback from the three groups was organized into 11 categories, and is shown in Table 8 with the top three categories for each group highlighted in bold.

Based on the interview results from the experimental group, three key advantages of the CM-GAI-CL mode were identified: "deepening learning content," "enhancing learning efficiency," and "improving the ability to integrate and verify information."

"Deepening learning content" was the most frequently mentioned, indicating the mode's effectiveness in terms of fostering deeper understanding. For example, E1–19 stated, "It helps me understand the topic better by providing more relevant examples and definitions quickly." E1–30 mentioned, "ChatGPT thinks more comprehensively than I do, identifying much knowledge I usually overlook. Concept mapping also helps me understand the learning materials." The visual nature of concept maps and immediate feedback from ChatGPT helped students grasp and retain content more intuitively. In terms of "enhancing learning efficiency," the experimental group reported a significant improvement in their learning efficiency. Concept mapping helped students quickly grasp key learning points, while ChatGPT's quick feedback accelerated their problem solving and boosted their overall efficiency. For instance, E1–32 expressed, "This diverse learning mode allows me to learn new knowledge. Also, ChatGPT is convenient, fast, and can provide more information." Similarly, E1–1 stated, "Using AI made the learning process more efficient, and a concept map clearly presented our learning outcomes. If there were any questions, we could ask ChatGPT." As for "improving the ability to integrate and verify information," this advantage applied solely to the experimental group, as only this group utilized the learning mode that combined concept mapping with ChatGPT. For example, E1–32 said, "Concept mapping helps me better understand the learning content, organize key points, and enhance my memory. Conventional methods are less effective, and this learning mode allows me to learn different knowledge." Concept mapping also supported students in organizing and integrating information more effectively. For instance, E1–6 mentioned, "Concept mapping helps me understand the content and clarifies my thought process." Also, E1–3 reported, "ChatGPT has extensive knowledge and can answer questions. Concept mapping can visualize the learning content, making the overall knowledge clearer."

According to the interview results from control group 1, three key advantages of the GAI-CL mode were identified: "deepening learning content," "enhancing learning efficiency," and "expanding diverse perspectives."

For example, E2–21 stated, "We rarely search for answers on our own. I enjoy this learning mode, as it helps me better understand the knowledge." Although control group 1 mentioned "deepening learning content" less frequently than the experimental group, their responses still reflected an emphasis on understanding the materials. The absence of concept mapping may have limited their depth of engagement compared to the experimental group. In terms of "enhancing learning efficiency," 10 students in control group 1 experienced improvements in their learning efficiency. For example, E2–23 stated, "Using ChatGPT to search for information is easier and more convenient." Similarly, E2–22 commented, "Interacting with ChatGPT is faster and allows me to test my understanding simultaneously." However, due to fewer structured strategies, students may have struggled to fully organize the information retrieved from ChatGPT. As for "expanding diverse perspectives," seven students in control group 1 believed that participating in the activity

Table 7
ANCOVA results of problem-solving skills.

Group	N	Mean	SD	Adjusted Mean	F	p	η^2	Post Hoc (LSD)
(E) Experimental	33	4.19	0.44	4.21	5.18	$p < .001$	0.10	E > C2
(C1) Control 1	26	3.89	0.72	3.82				E > C1
(C2) Control 2	29	3.78	0.68	3.82				

Table 8

Themes, codes, and numbers of occurrences of the interview results.

Theme	Code	The number of times mentioned		
		E	C1	C2
Deepening learning content and enhancing learning efficiency	Deepening learning content	24	12	9
	Enhancing learning efficiency	21	10	3
Promoting higher-order thinking	Improving the ability to integrate and verify information	15	-	-
	Promoting critical thinking	7	4	2
	Fostering problem-solving skills	9	5	2
	Strengthening questioning skills	12	1	0
	Expanding diverse perspectives	11	7	5
Strengthening collaboration and communication skills	Enhancing communication skills	7	1	3
	Facilitating collaborative learning	9	4	4
Providing immediate feedback and boosting learning motivation	Providing immediate feedback	7	3	1
	Achieving a sense of accomplishment	6	2	0

helped them broaden their thinking from different perspectives. For instance, E2-01 conveyed, “Through discussions with ChatGPT and my peers, I was able to draw more conclusions from different perspectives.” This suggested that students benefited from exposure to diverse viewpoints during the course, especially due to ChatGPT-assisted discussions and education for sustainable development (ESD)-related tasks.

Based on the interview results from control group 2, three key advantages of the C-CL mode were identified: “deepening learning content,” “expanding diverse perspectives,” and “facilitating collaborative learning.”

Regarding “deepening learning content,” control group 2 reported fewer instances of enhanced learning depth compared to the other two groups. Their reliance on teacher-guided instruction may have limited student motivation for deep engagement. As for “expanding diverse perspectives,” control group 2 mentioned this less frequently. In the C-CL mode, students primarily received viewpoints from the teacher, peers, and online searches, resulting in fewer opportunities to explore multiple viewpoints. In terms of “facilitating collaborative learning,” four students in control group 2 experienced benefits from the course design, which included group activities and discussions. Although the application of collaborative learning for this group was not as extensive as in the other two groups, it still fostered peer support and knowledge sharing.

Based on the interview results, most students in the experimental group identified three main advantages of the CM-GAI-CL mode: “deepening learning content” (24 students), “enhancing learning efficiency” (21 students), and “improving the ability to integrate and verify information” (15 students). On the other hand, students in control group 1 generally recognized the three advantages of the GAI-CL mode: “deepening learning content” (12 students), “enhancing learning efficiency” (10 students), and “expanding diverse perspectives” (7 students). In addition, students in control group 2 reported three advantages of the C-CL mode: “deepening learning content” (9 students), “expanding diverse perspectives” (5 students), and “facilitating collaborative learning” (4 students).

5. Discussion and conclusions

5.1. Discussion

This study investigated the impact of the CM-GAI-CL mode on students’ learning achievement in ESD, collective efficacy, critical thinking tendencies, and problem-solving skills. The results indicated that the experimental group outperformed control groups 1 and 2 across all aspects. The research questions are discussed and conclusions presented as follows. Overall, these findings provide empirical support for the cognitive-motivational pathway proposed in this study. Consistent with Social Cognitive Theory (Bandura, 1986, 2000), the findings indicate that structured cognitive scaffolding and collaborative interaction strengthened learners’ collective efficacy, thereby facilitating strategic engagement in complex tasks. This engagement, in turn, supported the development of higher-order thinking processes, including critical thinking and problem solving, ultimately contributing to improved learning achievement and more positive learning experiences.

Regarding the first research question, the results indicated that the CM-GAI-CL mode had a significant impact on students’ learning achievement. In this collaborative learning mode, GAI served as a personalized learning guide, while concept mapping was utilized as a strategy to help students organize the information or suggestions provided by GAI. The experimental results suggest that GAI alone might not significantly enhance learning achievement, which could be due to the metacognitive laziness induced by using GAI as indicated by some researchers (Fan et al., 2025). This phenomenon manifests when learners overly rely on GAI tools to perform cognitive tasks without critical reflection on the process or the validity, appropriateness, and accuracy of the GAI output. Although no significant difference was found in the “understanding” dimension across the three learning modes, the qualitative patterns and learning behaviors observed in the experimental group suggest that concept mapping may facilitate the process of organizing and interpreting the information provided by GAI. This aligns with Chang et al. (2022) and Lin et al. (2023), who indicated that concept mapping can support learners in grasping the overall structure of complex knowledge, even when such effects might not immediately reveal statistically significant performance differences. Finally, ethical considerations remain salient in AI-assisted learning. In particular, AI bias, misinformation, and student data privacy may shape learners’ experiences and outcomes, underscoring the need for

instructional guidance, monitoring, and explicit training in verification practices during AI-supported activities (Barzilai et al., 2025). For example, even when concept mapping is used to help learners structure and integrate GAI-generated ideas, students' understanding may still be constrained if the content being organized is inaccurate or misleading and learners do not consistently apply verification strategies.

Regarding the "remembering" items, the experimental group and control group 2 outperformed control group 1. The experimental group established connections between knowledge and information using graphical and visual concept maps, which significantly enhanced memory retention (Machado & Carvalho, 2020; Shi et al., 2023). As for control group 2, familiarity with the learning mode allowed students to engage with the content more quickly, reducing the time and effort required to adapt to new technologies, thereby positively influencing their retention. While ChatGPT provided rich conversational interactions for control group 1, this mode of interaction might have increased students' cognitive load, particularly when they needed to process large amounts of information (Dwivedi et al., 2023). The "remembering" items typically required students to recall extensive, specific knowledge quickly and effectively. Excessive reliance on conversations might divert students' attention, thereby hindering their memory performance. With regard to the "applying and analyzing" items, the experimental group significantly outperformed the other two groups. These items typically required students to apply learned knowledge to new situations or solve real-world problems, and the use of concept mapping helped students better integrate and apply knowledge, thereby enhancing their application skills (Evans & Jeong, 2023; Machado & Carvalho, 2020). The incorporation of ChatGPT also provided richer interaction and immediate feedback, fostering students' higher-order thinking and problem-solving skills (Chaunce & McKenna, 2023).

In addition, an analysis of the "applying and analyzing" items revealed that 11 students (33%) from the experimental group, six (23%) from control group 1, and four (14%) from control group 2 revised their selection of the sunset industry. These figures demonstrate the level of student engagement and reflection. Notably, the CM-GAI-CL mode not only increased the number of student responses, but also fostered more active participation and deeper thinking in order to understand and analyze issues related to sunset industries. Concept mapping allowed for the visualization and organization of complex information, helping students effectively integrate new and prior knowledge. This promoted a deeper understanding of the knowledge, and enhanced students' application skills (Evans & Jeong, 2023; Shi et al., 2023). Concept mapping may also have affected performance through reduced cognitive load and improved collaboration quality. As an external representation, a concept map can lower extraneous working-memory demands during complex learning by organizing and stabilizing information, thereby freeing up resources for integration and reasoning (Sweller, 1988). At the group level, the shared map functions as a common reference point that helps coordinate attention, negotiate meaning, and ground discussion, thereby supporting structured interaction and joint problem solving (Cicognani, 2000; Simone et al., 2001). In contrast, while ChatGPT possessed advantages such as generating diverse perspectives and providing rapid responses, its effectiveness was limited without additional supporting tools. Simply relying on ChatGPT for information might not result in significant improvements in students' learning achievement. When faced with large volumes of information, students might experience information overload and increased cognitive load. Therefore, this study highlights the necessity of integrating GAI with complementary learning strategies and technological tools. Such integration is crucial for realizing the full potential of GAI, promoting deep learning and independent thinking throughout the knowledge construction process.

With regard to the second research question, the results suggest that the experimental group demonstrated significantly higher collective efficacy compared to control groups 1 and 2. Collective efficacy plays a critical role in the collaborative learning process (Bandura et al., 1999; Park & Shin, 2022). The findings aligned with those of Chang et al. (2022) and Ma (2023), implying that the use of concept mapping positively influenced the collective efficacy. First, concept mapping enabled students to track their learning progress, thereby enhancing their confidence in completing tasks. Second, concept mapping assisted students in organizing and managing information systematically, increasing their sense of control and boosting confidence in their abilities (Park & Shin, 2022). More importantly, the creation of concept maps in this study required collaborative discussions and knowledge co-construction among students. This process enhanced students' self-efficacy in the group, thereby facilitating more effective collaborative learning (Shi et al., 2023; Tsourela et al., 2015). These findings suggest a potential benefit of CM-GAI-CL, although alternative explanations such as novelty effects or teacher feedback may also contribute.

As for the third research question, the findings indicated that the experimental group had significantly higher critical thinking tendencies compared to control groups 1 and 2. In this study, interacting with ChatGPT enabled students to acquire information; however, they were required to evaluate its accuracy and applicability. This learning process contributed to the development of their critical thinking abilities (Huang et al., 2017; Rudolph et al., 2023). Moreover, constructing concept maps during group discussions provided students with opportunities to verify information. By organizing collected information and connecting relationships between different concepts within the concept map, students evaluated its accuracy and reliability, thus fostering their critical thinking skills (Barta et al., 2022). The process of collaboratively analyzing problems and proposing solutions during group discussions further enhanced students' critical thinking and problem-solving skills (Knoche, 2022; Quines, 2017).

In response to the fourth research question, the findings showed that the experimental group demonstrated significantly better problem-solving skills compared to control groups 1 and 2, influenced by the use of both concept mapping and ChatGPT. First, concept mapping provided a systematic analytical framework for students to identify and understand problems more clearly (Barta et al., 2022; Shi et al., 2023). Through concept mapping, students could break down complex issues into smaller components, making it easier to figure out solutions (Evans & Jeong, 2023; Tsai & Wu, 2021). Meanwhile, ChatGPT offered immediate answers and suggestions, supporting students in refining their thinking rather than simply providing direct answers (Chauncey & McKenna, 2023; Dwivedi et al., 2023). By incorporating ChatGPT and concept mapping, students were able to integrate information from multiple perspectives, which enhanced their analytical and problem-solving skills. This improvement reflects students' active engagement in questioning, evaluating, and integrating AI-generated responses, rather than passive reliance on AI outputs. These results were in line with the interview

results. For example, E1–19 stated that to tackle the problem, she must understand how to ask ChatGPT to get a more complete and accurate answer. Concept mapping was very helpful for organizing problems and identifying key points. Thus, it was concluded that the integration of concept mapping and ChatGPT contributed to enhancing students' analytical and problem-solving skills, ultimately promoting their learning achievement and effectiveness.

Regarding the fifth research question, the findings indicated that the experimental group outperformed both control groups 1 and 2 in most aspects, particularly in deepening learning content, enhancing learning efficiency, and improving the ability to integrate and verify information. Further analysis revealed commonalities among the three groups in “deepening learning content,” “expanding multiple perspectives,” and “promoting collaborative learning.”

During the interviews, all three groups mentioned “deepening learning content.” In collaborative learning, students typically engaged in-depth discussions, exchanges, and knowledge sharing, which fostered a deeper understanding of the learning content (Rorimpandey et al., 2022; Suendarti & Virgana, 2022). As for “expanding diverse perspectives,” all three learning modes helped students broaden their perspectives. The core of collaborative learning was the interaction between students, which allowed them to learn the viewpoints and ideas of their peers, thereby expanding their thinking angles (Gillies, 2003; Hsu et al., 2023). In addition, all three learning modes were reported to “promote collaborative learning.” The core principle of collaborative learning, regardless of technological assistance, was to encourage collaboration and interaction among students (Knoche, 2022; Suendarti & Virgana, 2022). Johnson and Johnson (2009) pinpointed that collaborative learning promoted positive interdependence among students, and helped them achieve shared learning goals through teamwork. As a result, regardless of the learning mode adopted, this study revealed the importance of collaboration in the learning process.

Based on the interview results, three key differences were identified among the three groups: “improving the ability to integrate and verify information,” “strengthening questioning skills,” and “achieving a sense of accomplishment.” Regarding “the ability to integrate and verify information,” the experimental group showed significant improvement. This improvement was likely due to concept mapping, which helped students structure knowledge and identify its logical connections (Evans & Jeong, 2023; Machado & Carvalho, 2020). Additionally, GAI provided immediate knowledge and verification, allowing students to better organize and examine the knowledge they had acquired (Kasneji et al., 2023; Rudolph et al., 2023). On the contrary, for control group 1, the lack of an effective learning tool could limit the full potential of GAI. Similarly, control group 2 relied entirely on the conventional collaborative learning mode, resulting in no significant progress in information integration and verification.

Regarding the theme of “promoting questioning skills” that emerged from the student interviews, more students in the experimental group referred to this perspective compared to those in the control groups. As noted by Chin and Brown (2000), the development of questioning skills is often associated with students' deeper understanding of subject matter. In the experimental group, the use of concept mapping likely facilitated the organization of knowledge structures, enabling students to identify gaps, inconsistencies, or areas of uncertainty in their understanding. This cognitive process, in turn, encouraged them to ask more questions.

In terms of “achieving a sense of accomplishment,” concept mapping assisted students in establishing clearer knowledge structures during their learning process. When students could link various types of knowledge and key points together, it enhanced their self-efficacy, thereby increasing their sense of accomplishment (Evans & Jeong, 2023; Ma et al., 2023). Although control group 1 also used ChatGPT, students struggled with understanding and applying the learning content, leading to a lower sense of accomplishment. Control group 2, which relied entirely on the conventional collaborative learning mode, also encountered more challenges, making it harder for them to experience a sense of accomplishment. According to Bandura's (1999) self-efficacy theory, when students experience success in the learning process, their motivation and sense of achievement are enhanced. Therefore, the learning mode adopted by the experimental group was more effective in terms of increasing students' sense of accomplishment. In addition, the CM-GAI-CL mode helped students organize their knowledge structure by building conceptual maps through repeated ChatGPT question-and-answer sessions, instant solutions, and feedback. This process resulted in a number of small but concrete mastery experiences, which helped students increase their self-efficacy, leading to a sense of accomplishment (Bandura, 1999). In this study, the experimental group students also frequently mentioned feelings such as “being more confident in completing tasks” and “feeling more capable than before” in their interviews, which echoes Evans and Jeong (2023) and Ma et al. (2023), who emphasized that structured learning tools paired with synergistic learning strategies can enhance learners' self-efficacy and academic achievement. The observed improvements in the experimental group may be explained by several underlying psychological mechanisms. First, increased self-efficacy and collaborative engagement likely enhanced students' motivation and participation. Concept mapping and GAI facilitated students' sense of control over their learning, enabling them to organize information, track progress, and gain confidence during collaborative problem-solving tasks. Second, reduced cognitive load and improved metacognitive regulation allowed students to focus on deeper understanding and critical evaluation. By organizing information visually through concept maps and verifying GAI outputs, students could engage in higher-order cognitive processes such as analysis and synthesis. Third, students' sense of accomplishment and intrinsic motivation likely contributed to their superior outcomes. The immediate feedback from GAI, combined with structured concept mapping, provided mastery experiences and positive reinforcement, enhancing engagement, persistence, and the likelihood of transferring learned skills to new problem-solving tasks.

5.2. Limitations and future studies

The main limitation of this study was its short duration, which may have impacted the effectiveness of the concept mapping and use of GAI, particularly for assessing higher-order skills such as critical thinking tendencies. Future researchers are encouraged to extend the experimental period for a more comprehensive analysis of various aspects. Additionally, conducting a longitudinal study to observe the sustained impact of concept mapping and GAI on students' learning achievement would provide valuable insights into the role of

these technological tools in students' learning processes.

Moreover, this study identified three major limitations in the application of GAI in education. First, due to its limited semantic understanding, GAI sometimes failed to fully grasp the context or complex questions, leading to irrelevant or inaccurate responses. Second, this study used ChatGPT (GPT-3.5 version) because most students could not afford to purchase GPT-4.o. Since ChatGPT was primarily used for discussion and consultation during the learning activities, the selected version was adequate to support the intended learning design. However, adopting a more advanced version such as GPT-4.o in future studies may enable more diverse and sophisticated learning designs by offering improved contextual understanding, reasoning capabilities, and personalized support. Third, GAI lacks emotional or human perceptive abilities, preventing it from truly understanding or empathizing with users' emotions and circumstances. It might also generate responses containing biases or inappropriate content, requiring students to critically evaluate and filter the information. While the teacher guided students in verifying the information provided by GAI, it is important to note that other search engines could also present incorrect information. Hence, cultivating students' critical thinking and information filtering skills is crucial, not only for using GAI but also for various other information retrieval methods.

Lastly, we propose the following recommendations to guide future research in exploring the challenges and potential of AI tools in education:

1. It is recommended that future research explore how to integrate GAI with other personalized learning systems or to develop further personalized AI models that can automatically adjust content and teaching methods based on students' learning situations and performance, thereby enhancing the educational applicability of AI tools. In this context, concept mapping can be incorporated as a visual and diagnostic tool to represent students' cognitive structures, enabling the AI system to better identify learning gaps and tailor instructional prompts accordingly.
2. Future research should integrate GAI with various teaching modes, using it as a supplementary tool rather than as the primary learning pathway. Educators could combine AI technology with conventional teaching methods to foster collaboration and discussion between teachers and students, thereby balancing the importance of technological applications with interpersonal communication.
3. The convenience of immediate feedback provided by GAI may reduce opportunities for students to engage in independent thinking. If students become accustomed to relying on GAI for answers, they may neglect the underlying logical reasoning and analytical processes, potentially leading to a decline in their cognitive abilities. Therefore, it is recommended that future educators design challenging tasks that require students to reflect, summarize, and critique after using AI. For instance, students could be asked to further explain or expand upon the AI-generated answers, ensuring that they not only understand the results but also grasp the underlying processes and principles.
4. Since GAI's responses are based on existing data and patterns, it may not be effective in terms of fostering students' creative thinking, particularly in situations that require divergent thinking or innovative problem solving. Future research could explore how to integrate GAI with creative learning activities, such as designing tasks that ask students to critique or expand on AI-generated answers, thereby stimulating their creative thinking. Additionally, teachers could use GAI to generate different options or perspectives, prompting students to engage in assessments, make comparisons, and express their creativity.
5. It is also crucial to consider ethical issues, including AI bias, misinformation, and student data privacy.
6. In addition, exploring the role of teacher facilitation, cooperative learning dimensions, and the relationship between affective and metacognitive constructs may further enhance the educational effectiveness of AI-assisted learning.

The limitations and recommendations presented in this study not only highlight the technical and social challenges of applying ChatGPT in education, but also provide innovative perspectives on how to more effectively integrate AI into educational systems. In addition, the CM-GAI-CL model developed for this study demonstrated strong theoretical and practical contributions by innovatively combining concept maps and GAI in the ESD curriculum. At the theoretical level, this study breaks through the single application of GAI tools in language learning or writing assistance, and fills the gap in the literature on how AI can support sustainability, systems thinking, and civic responsibility education. At the practical level, the CM-GAI-CL model provides educators with an operational and flexible teaching framework to help students engage in interdisciplinary thinking, critical analysis, and civic engagement when faced with authentic issues. This not only enhances students' understanding of sustainable development, but also provides a feasible template for future AI-assisted sustainable education curriculum design, which is of far-reaching significance to the development of education technology.

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Ethics approval

The ethical requirements for research in this selected university were followed.

CRedit authorship contribution statement

Shih-Ting Chu: Writing – review & editing, Writing – original draft, Visualization, Validation, Project administration,

Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Gwo-Jen Hwang:** Writing – review & editing, Writing – original draft, Visualization, Validation, Project administration, Methodology, Investigation, Conceptualization. **Yun-Fang Tu:** Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Conceptualization.

Declaration of competing interest

The authors declare no conflicts of interest.

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Appendix A. Coding scheme for content analysis (revising the sunset industry they selected after a group discussion)

Category	Sub-category
A1 Final selection of the sunset industry	A1-01: Manufacturing A1-02: Construction A1-03: Arts, Entertainment, and Recreation A1-04: Education A1-05: Accommodation and Food Service Activities A1-06: Services and Sales A1-07: Professional, Scientific, and Technical Activities A1-08: Information and Communication Equipment
A2 Reasons for the disappearance of the industry	A2-01: Rising costs A2-02: Advancements in technology and science A2-03: Declining birthrate A2-04: Lack of successors A2-05: Low wages A2-06: Decreased market demand A2-07: Changes in consumer behavior A2-08: Emergence of substitutes A2-09: Pressures on environmental sustainability A2-10: Higher risk/Greater effort
A3 Measures to improve or delay the disappearance of the industry	A3-01: Support from government policy A3-02: Training for skill enhancement (Talent development) A3-03: Integration with tourism A3-04: Incorporation into educational materials A3-05: Celebrity endorsement/Advertising promotion A3-06: Application/Transformation through new technologies A3-07: Wage increases A3-08: Support for environmental sustainability A3-09: Cost reduction A3-10: Adjustment of marketing strategies A3-11: Organizing related competitions
A4 New facts or perspectives learned in group discussions	A4-01: The importance of sunset industries A4-02: Specific improvement measures for sunset industries A4-03: Reasons for the disappearance of sunset industries A4-04: Changes in market demand A4-05: Cost A4-06: Labor shortage A4-07: Environmental trends

Data availability

The data and materials are available upon request to the corresponding author.

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